$$A+B \rightleftharpoons C + D$$

The free energy change of the above reaction in any state (ΔG) is related to the standard free energy change of the reaction (ΔG^0) according to the following equation.

$$\Delta G = \Delta G^0 + RT \ln Q \text{ ------(7.39)}$$

where Q is reaction quotient and is defined as the ratio of concentration of the products to the concentrations of the reactants under non equilibrium condition.

When equilibrium is attained, there is no further free energy change i.e. $\Delta G = 0$ and Q becomes equal to equilibrium constant. Hence the above equation becomes.

$$\Delta G^0 = -RT \ln K_{eq}$$

This equation is known as Van't Hoff equation.

$$\Delta G^0 = -2.303\ RT \log K_{eq} \text{ ----------(7.40)}$$

We also know that

$$\Delta G^0 = \Delta H^0 - T\ \Delta S^0 = -\ RT \ln K_{eq}$$

Problem: 7.9

Calculate ΔG^0 for conversion of oxygen to ozone $3/2\ O_2 \rightleftharpoons O_{3(g)}$ at 298 K, if K_p for this conversion is 2.47×10^{-29} in standard pressure units.

Solution:

$$\Delta G^0 = -\ 2.303\ RT \log K_p$$

Where

$R = 8.314\ JK^{-1}mol^{-1}$

$K_p = 2.47 \times 10^{-29}$

$T = 298K$

$\Delta G^0 = -2.303(8.314)(298)\log(2.47\times10^{-29})$

$\Delta G^0 = 163229\ Jmol^{-1}$

$\Delta G^0 = 163.229\ KJ\ mol^{-1}$

7.12 Third law of Thermodynamics

The entropy of a substance varies directly with temperature. Lower the temperature, lower is the entropy. For example, water above 100^0 C at one atmosphere exists as a gas and has higher entropy (higher disorder). The water molecules are free to roam about in the entire container. When the system is cooled, the water vapour condenses to form a liquid. The water molecules in liquid phase still can move about somewhat freely. Thus the entropy of the system has decreased. On further cooling, water freezes to form ice crystal. The water molecules in the ice crystal are highly ordered and entropy of the system is very low.

If we cool the solid crystal still further, the vibration of molecules held in the crystal lattice gets slower and they have very little freedom of movement (very little disorder) and hence very small entropy.

At absolute zero {0 K (or) -273^0C}, theoretically all modes of motion stop.

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